



The Intrusiveness Paradox: A Systematic Review of the Dark Side of Artificial Intelligence in Consumer Behavior

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Abstract: This study examines the emerging dark side of Artificial Intelligence (AI) in marketing by addressing how AI-driven personalization shapes consumer perceptions of intrusiveness, privacy, and trust. While AI enhances personalization and customer experience, it simultaneously raises concerns about surveillance and loss of autonomy, creating a fundamental tension referred to as the Intrusiveness Paradox. To investigate this issue, a systematic literature review (SLR) was conducted following established review protocols, analyzing 56 peer-reviewed journal articles published between 2022 and 2026. The study combines bibliometric mapping with a structured synthesis framework to identify dominant themes, theoretical foundations, research contexts, and methodological patterns. The findings reveal three primary research streams: emotional and psychological drivers such as perceived creepiness and human-like system design; the trade-off between privacy concerns and trust in data-driven personalization; and the impact of these factors on marketing outcomes including customer experience and brand attitudes. The results show that increased personalization and anthropomorphic design often intensify perceptions of surveillance, reduce trust, and trigger resistance among consumers. Despite rapid growth in this field, literature remains fragmented and heavily reliant on short-term and experimental approaches, with limited attention to longitudinal and real-world contexts. The study concludes that the negative consequences of AI are not isolated effects but interconnected responses reflecting a deeper tension between personalization and autonomy. By integrating these perspectives, the study contributes a unified conceptual understanding of the Intrusiveness Paradox and highlights the importance of transparent, ethical, and balanced AI design. These insights provide guidance for both researchers and practitioners seeking to develop AI systems that enhance value while preserving consumer trust and autonomy.

Keywords: Intrusiveness Paradox; Artificial Intelligence; Consumer creepiness; Systematic literature review; Dark side of AI; Privacy paradox; Uncanny valley; Consumer behavior

JEL Classification: M31, M15, D91, D83, O33

1. Introduction

Artificial Intelligence (AI) has transformed contemporary marketing by enabling organizations to personalize customer experiences, automate decision-making, and predict consumer preferences with unprecedented accuracy. AI-driven applications such as recommendation systems, conversational agents, and personalized advertising have become integral components of digital marketing strategies. While these technologies create substantial value for firms and consumers, they simultaneously raise concerns regarding privacy, perceived surveillance, algorithmic transparency, and consumer autonomy. Customers might wonder how AI systems possess extensive personal information. This tension can lead to the Intrusiveness Paradox.

AI improves personalization, but it also makes people feel like they're being watched and have less control. This paradox gets stronger as AI systems become more like people and better at making predictions. AI is now a big part of modern marketing. Firms use chatbots, algorithms, and virtual agents daily (Chandra et al., 2022; Mou & Meng, 2024). These systems learn from data and make decisions. They personalize offers and predict consumer

needs (Saura et al., 2024). AI improves speed and relevance. It also affects how people think and feel. Some users feel uneasy or as if they are being watched. Others question how much AI knows about them (Gaczek et al., 2023; Lv & Huang, 2022). This reaction is often called consumer creepiness. Creepiness is a sense of discomfort or threat. It appears when AI feels too human or too intrusive (Chaturvedi et al., 2023). It also appears when firms hide how systems work (Keegan et al., 2023). Prior research studies privacy, trust, or anthropomorphism alone. Many studies focus on the benefits of AI and performance gains (Labrecque et al., 2024). Fewer examine harmful side effects in a unified way. Findings remain scattered across theories and contexts. Recent studies indicate that the dark side of AI marketing extends beyond privacy concerns to include psychological manipulation, financial vulnerability, and broader ethical challenges arising from increasingly sophisticated algorithmic persuasion. These emerging concerns reinforce the need for an integrative framework capable of explaining the multiple adverse consequences of AI-enabled marketing rather than examining each issue in isolation (Shahab et al., 2025). Recent scholarship has similarly emphasized that the growing adoption of AI in marketing creates not only opportunities for improved decision-making and customer engagement but also significant risks related to trust, ethics, transparency, and unintended consumer consequences, highlighting the need for more comprehensive investigations into the darker side of AI-enabled marketing (Pantano et al., 2023).

Although recent evidence has synthesized the negative consequences of AI in marketing (Barari et al., 2024), existing reviews primarily catalogue adverse outcomes without offering an integrative theoretical framework explaining how the benefits of AI-enabled personalization coexist with consumers' perceptions of psychological intrusion. Similarly, recent bibliometric evidence has mapped the growing body of research on the dark side of AI in consumer behavior, highlighting increasing scholarly attention to privacy concerns, algorithmic bias, consumer trust, and ethical challenges. However, these studies primarily describe research trends and intellectual structures rather than developing an integrated conceptual explanation of the underlying mechanisms linking these adverse outcomes (Muslik et al., 2025). This review argues that these effects reflect one deeper tension. We call this the Intrusiveness Paradox. The same AI tools that improve personalization also raise fear. Greater relevance can increase perceived surveillance (Saura et al., 2024). Human-like design can boost warmth but later cause discomfort (Vo et al., 2023). Uncanny Valley Theory accounts for increasing discomfort at high levels of realism (Chen et al., 2023). Privacy calculus explains data trade-offs (Gong et al., 2023). psychological reactance explains why people resist being controlled. However, these theories usually treat these outcomes as separate issues.

Current research does not comprehensively address the core issue of perceived autonomy loss. As AI is increasingly integrated into user experiences, individuals report diminished control. This tension is central to the risks associated with AI in marketing. To date, no study has systematically mapped this tension across the field, nor has any review synthesized these topics into a consolidated framework. Consequently, a structured synthesis is required.

Organizations are gradually deploying AI yet often provide limited transparency. Opaque systems diminish trust and increase skepticism (Cloarec et al., 2024). Perceptions of surveillance can diminish brand value. Managers need clearer, research-based guidance. This study performs a systematic literature review (SLR) following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines to reduce bias. The adoption of a SLR is consistent with recent review studies conducted across different disciplines, where SLR methodology has been employed to synthesize fragmented bodies of knowledge, identify research trends, and establish future research agendas (Riandhi et al., 2025). The review encompasses 56 peer-reviewed articles published between 2022 and 2026. The TCCM (Theory, Context, Characteristics, Methodology) framework is applied to systematically organize the findings.

Beyond synthesizing the existing literature, this review develops and theoretically positions the Intrusiveness Paradox as an integrative framework that explains how the benefits of AI-enabled personalization coexist with consumers' perceptions of psychological intrusion, thereby extending existing theories of privacy, trust, and consumer behavior.

1.1 Research Questions

This review addresses four research questions:

RQ1: What themes shape research on the dark side of AI?

RQ2: Which theories explain AI creepiness and intrusiveness?

RQ3: What contexts, constructs, and methods dominate this field?

RQ4: Where do gaps remain for future research?

This paper reframes the dark side of AI, stressing the connections among creepiness, privacy apprehensions, and the loss of trust. These components together illustrate a paradox inherent in AI-based personalization, wherein AI systems are expected to increase relevance while preserving individual dignity. This review presents the AI Intrusiveness Paradox, which explains how the same AI mechanisms that improve personalization may also lead to perceptions of intrusion and psychological discomfort.

This SLR is organized according to the Introduction, Methods, Results, and Discussion structure. The following

sections include the literature review, data and methods, results, discussion and limitations, and finally, conclusions and future directions.

1.2 Literature Review

1.2.1 AI in marketing

AI means machines can do tasks that usually need human intelligence, like learning, predicting, language processing, and making decisions. In marketing, AI helps analyze large-scale customer data and is used for things like product recommendations, chatbots, targeted advertising, and automating services. Early AI mostly made operations faster and more efficient, but today's AI can also interact with customers using voice, text, and even human-like avatars. Considering these interactive capabilities, many scholars describe AI as a social actor in market exchanges (Chandra et al., 2022; Saura et al., 2024). Consumers often respond to AI systems as if they were human agents, demonstrating trust, engagement, or emotional discomfort (Cloarec et al., 2024; Gutuleac et al., 2024).

AI improves service speed, personalization, and customer convenience, leading to large-scale adoption by firms hoping to boost customer experience and marketing outcomes. Personalization has become a strategic capability that enables organizations to tailor products, services, and marketing communications to individual consumer preferences, thereby enhancing customer engagement, customer experience, and perceived value (Kaushik & Sharma, 2023; Kumar, 2024). However, these systems may likewise provoke consumer unease. Customers sometimes perceive that they are being monitored, analyzed, or manipulated (Chen et al., 2023). This tension points to the potential negative consequences, often referred to as the dark side of AI in marketing.

1.2.2 The dark side of AI in marketing

Early marketing research mainly focused on the benefits of digital tools, showcasing efficiency, personalization, and improved service quality. However, recent studies indicate that technology may also create negative consumer responses. These reactions include privacy concern, distrust, fatigue, and perceived manipulation (Cloarec et al., 2024; Saura et al., 2024). Similar concerns have also been identified in broader digital environments, where algorithmic interactions across online platforms and social media may undermine consumer trust, transparency, and overall well-being (Dogru et al., 2025). AI systems depend heavily on personal data to predict consumer actions. Consumers may worry about how firms collect and use this information.

Highly personalized messages can be viewed as intrusive or excessively familiar. Consumers may question how firms know so much about them. Researchers describe these reactions as the dark side of AI-enabled marketing. This concept refers to negative psychological or behavioral responses to AI systems, which may diminish trust, satisfaction, and long-term brand loyalty (Chandra et al., 2022).

1.2.3 Consumer creepiness

One key reaction in this area is consumer creepiness. Creepiness refers to a feeling of unease when people face an unclear social threat. The threat feels ambiguous rather than clearly dangerous. People often feel creeped out when technology behaves almost like a human. The system appears too human while remaining recognizably artificial. This reaction also arises when organizations know unexpected information about customers. Consumers may wonder how the firm obtained such information (Gutuleac et al., 2024).

Creepiness frequently encompasses perceptions of surveillance and diminished autonomy. Consumers may perceive covert data tracking or algorithm-based monitoring, leading to avoidance behaviors. Consumers may discontinue use of a service or reject automated interactions. Understanding creepiness is necessary for organizations to implement AI-driven marketing systems.

1.3 Theoretical Background

Several behavioral theories help explain consumer reactions to AI systems. Three theories appear often in this research area. These theories explain how consumers interpret AI interactions and how they perceive intrusion.

1.3.1 Uncanny Valley Theory

Uncanny Valley Theory describes human responses to technology with human-like features. The theory proposes that individuals tend to prefer machines that exhibit moderate human resemblance, as this level of likeness enhances familiarity and emotional comfort. However, when machines become nearly indistinguishable from humans, people often feel discomfort instead of attraction. This decline in comfort is referred to as the uncanny valley.

Human-like chatbots, digital assistants, and virtual influencers may trigger this reaction. Consumers may feel uneasy when machines mimic human emotions too closely. In marketing, this excessive similarity can reduce trust, leading consumers to question the authenticity of the interaction (Saura et al., 2024). Uncanny Valley Theory helps

explain why AI sometimes feels creepy rather than helpful.

1.3.2 Privacy calculus theory

Privacy calculus theory describes how consumers weigh the benefits and risks of disclosing personal data. Benefits such as convenience, speed, and personalization are weighed against risks such as data misuse or unwanted tracking. Consumers determine if the advantages outweigh the risks. AI marketing systems depend on access to considerable amounts of personal data. Effective personalization relies on collecting detailed information about consumer behavior.

When consumers perceive significant benefits, they are more likely to accept data collection. Conversely, high perceived risks may lead to resistance (Cloarec et al., 2024). Privacy Calculus Theory highlights the persistent tension between personalization and privacy concerns.

1.3.3 Psychological reactance theory

Psychological Reactance Theory explains how people resist when they feel controlled. People value freedom when making decisions. They react negatively when that freedom seems restricted. Highly targeted marketing messages may diminish perceived autonomy. Consumers may feel manipulated or pressured. AI systems can also predict choices with high accuracy. This prediction may create feelings of surveillance or influence. Consumers may respond with avoidance, skepticism, or rejection. Reactance theory helps explain why intrusive personalization may harm brand trust. Recent evidence further suggests that AI-enabled marketing may evoke broader psychological responses beyond perceived loss of control. AI-induced existential threats can increase consumer skepticism toward corporate marketing communications by reducing confidence in AI-mediated interactions and organizational intentions. These findings reinforce the argument that psychological responses to AI extend beyond privacy concerns to include diminished trust and skepticism toward AI-enabled marketing communications (Yang et al., 2025).

1.3.4 Positioning the Intrusiveness Paradox

Unlike Privacy Paradox, which primarily explains why consumers continue to disclose personal information despite privacy concerns, the Intrusiveness Paradox explains the broader psychological tension created after AI-enabled personalization occurs. Rather than focusing solely on disclosure decisions, the proposed framework integrates emotional discomfort (consumer creepiness), perceived surveillance, autonomy loss, trust erosion, and resistance into a unified conceptual explanation. It therefore extends existing perspectives by demonstrating that these outcomes are interconnected manifestations of the same underlying paradox rather than independent phenomena.

Recent systematic review evidence further demonstrates that the Privacy Paradox is shaped by a complex interplay of cognitive, psychological, technological, and sociocultural factors that influence consumers' willingness to disclose personal information despite privacy concerns (Memarian et al., 2026). Building upon these insights, the Intrusiveness Paradox extends beyond disclosure behavior by incorporating perceived psychological intrusion, emotional discomfort, and consumer reactance to explain broader responses to AI-enabled personalization.

Table 1. Positioning the Intrusiveness Paradox

Existing Framework	Main Focus	Limitation	Contribution of Intrusiveness Paradox
Privacy paradox	Consumers disclose data despite privacy concerns	Focuses only on disclosure decisions	Explains emotional, cognitive and behavioral consequences after AI personalization
Privacy calculus	Cost-benefit evaluation	Rational decision model	Includes emotional discomfort and perceived autonomy loss
Consumer creepiness	Emotional discomfort	Explains only emotional response	Connects creepiness with trust erosion and resistance
Personalization-privacy trade-off	Balance between relevance and privacy	Mainly personalization outcomes	Explains why personalization simultaneously creates value and psychological intrusion
Intrusiveness Paradox (this study)	Simultaneous coexistence of personalization benefits and perceived surveillance, autonomy loss, and emotional discomfort	Integrates multiple theoretical mechanisms into one framework	-

Note: A hyphen (-) denotes not applicable; AI = Artificial Intelligence.

Similarly, while Privacy Calculus Theory assumes consumers evaluate benefits and risks through rational trade-offs, the Intrusiveness Paradox recognizes that AI interactions also generate affective responses that cannot be fully explained through economic reasoning. Human-like AI, algorithmic opacity, and perceived surveillance frequently trigger emotional reactions such as discomfort and psychological reactance even when personalization benefits remain high. Accordingly, the proposed framework complements rather than replaces existing theories by

integrating cognitive, emotional, and behavioral mechanisms within a single conceptual model (Table 1).

1.4 The Need for a Systematic Review

Recent years have witnessed a rapid expansion in research on AI in marketing. Numerous studies investigate AI chatbots, recommendation platforms, and automated services, while others address issues such as trust, privacy concerns, and technology acceptance. However, the existing literature remains fragmented. Existing studies utilize diverse theoretical frameworks, research contexts, and conceptual constructs. Recent scholarship further suggests that understanding AI-enabled marketing requires integrating AI capabilities with consumer psychology and marketing strategy, as consumer responses emerge from the interaction between technological functionality and psychological processes rather than from technological characteristics alone (Qureshi & Ahmed, 2026). However, their findings commonly address narrowly defined topics or focus on specific technologies. A comprehensive synthesis of the negative implications of AI for consumer behavior.

Existing reviews also focus mainly on AI adoption or technology acceptance; however, they rarely examine the psychological trigger of consumer creepiness. With recent advances in generative AI, there is an increased demand for updated analysis. As AI systems now appear more human and socially interactive, these changes intensify the tension between personalization and perceived intrusion. This tension reflects what can be described as the Intrusiveness Paradox. AI improves customer experience while also increasing perceptions of surveillance. A systematic review can clarify the main research patterns. It can also identify theoretical gaps and future research needs. To address this issue, the present study conducts a SLR.

The review follows the PRISMA framework to ensure transparency and reduce bias. The study analyzes recent literature using the TCCM framework. This approach maps dominant theories, contexts, constructs, and research methods.

1.5 Conceptual Model

The framework demonstrates the Intrusiveness Paradox in AI-enabled marketing. While AI technologies enhance personalization and efficiency, excessive anthropomorphism and data personalization may elicit feelings of creepiness and privacy concerns. Such responses can diminish trust and foster resistance to AI-driven interactions. Based on reviewed theories and constructs, this study introduces a conceptual model of the AI Intrusiveness Paradox. AI marketing systems rely on personalization and human-like interaction.

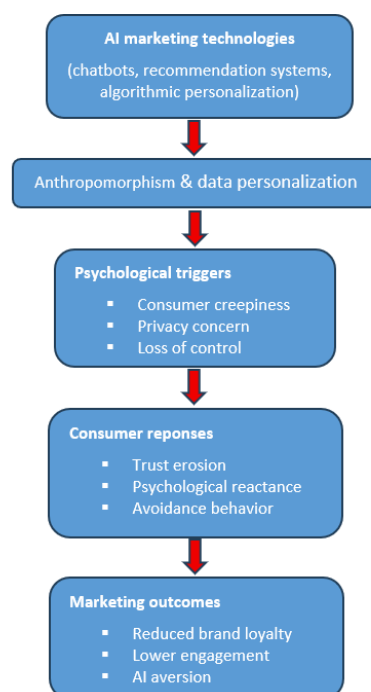


Figure 1. Conceptual framework of the AI Intrusiveness Paradox

Note: AI = Artificial Intelligence.

These features enhance efficiency and customer engagement; however, they also intensify perceptions of surveillance and psychological discomfort. These responses affect end-user trust, resistance, and the sustained

adoption of AI services. Figure 1 summarizes these conceptual relationships and guides the systematic review analysis.

The next section explains the methodology used to identify and analyze the 56 articles.

2. Methodology

2.1 Research Design

This study used a SLR. The review followed the PRISMA 2020 guidelines. PRISMA gives clear steps for selecting studies. It improves transparency and reduces bias. The goal of this paper is to identify and review research on the dark side of AI in marketing and consumer behavior.

2.2 Search Strategy

The literature search was conducted using Dimensions.ai, a database that indexes a wide range of academic journals. Dimensions.ai offers citation tracking and advanced filtering tools. Dimensions.ai was selected as the primary database because it provides broad interdisciplinary coverage across business, marketing, computer science, information systems, and the social sciences, making it particularly suitable for investigating the multifaceted nature of AI-enabled consumer behavior. Unlike databases that emphasize citation selectivity, Dimensions indexes a wider range of peer-reviewed journals while integrating citation metrics, funding information, and linked research outputs within a single platform. Since the objective of this review was to capture the emerging literature on the dark side of AI—a rapidly evolving topic spanning multiple disciplines—Dimensions offered broader coverage than would have been achieved through reliance on a single traditional citation database.

The search string was:

("Artificial Intelligence" OR "AI" OR "Chatbot" OR "Algorithm")

And

("Dark Side" OR "Creepiness" OR "Uncanny Valley" OR "Privacy Paradox" OR "Intrusiveness")

And

("Consumer Behavior" OR "Marketing" OR "Consumer Psychology")

To improve the comprehensiveness of the search, the search terms were developed iteratively through an initial scoping review of highly cited publications on AI marketing, consumer creepiness, privacy paradox, and anthropomorphic AI. Frequently occurring keywords, author keywords, and index terms appearing in seminal studies were examined and combined using Boolean operators (AND/OR). The preliminary search strategy was refined through several pilot searches until it consistently retrieved the principal publications in this research area.

As a validation step, the final search results were manually inspected to verify that influential publications identified during the preliminary scoping review were captured by the search strategy. Additional backward reference checking was also conducted for several highly cited review articles to ensure that no major streams of literature had been unintentionally omitted. The search was conducted in February 2026. It returned 1,530 records.

Nevertheless, the review has several methodological limitations. Although Dimensions.ai provides extensive interdisciplinary coverage, reliance on a single database may have excluded some relevant studies indexed exclusively in Web of Science, Scopus, or discipline-specific databases. Consequently, the findings should be interpreted as representing the dominant body of peer-reviewed literature rather than an exhaustive inventory of all published research. Future systematic reviews may benefit from combining multiple databases to further enhance coverage and reduce potential database-selection bias.

2.3 Selection Criteria

The following filters were applied to narrow the results.

2.3.1 Inclusion criteria

Only peer-reviewed journal articles were included. Publication years were limited to 2022–2026.

Only English-language studies were used. The following fields of research were selected:

- Marketing;
- Commerce, management, tourism and services;
- Business systems in context;
- Information and computing sciences;
- Commercial services;
- Strategy, management and organizational behavior;
- Tourism;

- Human-centered computing;
- Cybersecurity and privacy;
- Language, communication and culture;
- Philosophy and religious studies;
- Information systems;
- Library and information studies; and
- Applied ethics.

2.3.2 Exclusion criteria

- Conference papers were removed;
- Book chapters were removed;
- Editorials and notes were excluded;
- Industry reports were not included; and
- Studies outside marketing or consumer topics were excluded.

2.3.3 Screening and filtering process

The screening process followed PRISMA's four-stage procedure. The proposed screening and filtering process is illustrated in Figure 2.

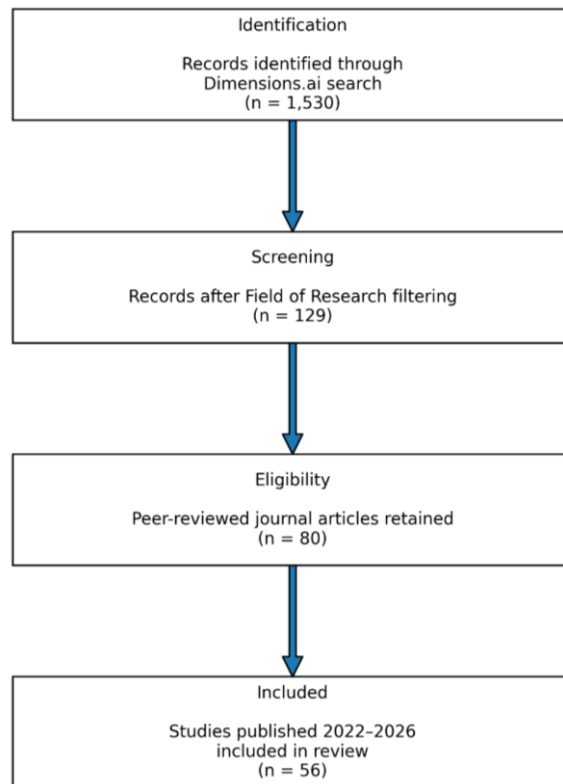


Figure 2. PRISMA 2020 flow diagram

Note: PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

These 56 articles formed the final dataset for bibliometric mapping and TCCM synthesis. Each article was reviewed carefully. Key details were extracted into a spreadsheet. The following data were recorded:

- Authors;
- Year;
- Theory used;
- Research context;
- Key constructs; and
- Method used.

The dataset was exported as a Comma-Separated Values (CSV) file from Dimensions. Bibliometric mapping techniques (co-occurrence and co-authorship analysis) were used to analyze patterns. A TCCM framework guided the final synthesis. This process ensured clarity and repeatability.

3. Results

3.1 Thematic Analysis Introduction

Following the geographical mapping, a thematic co-occurrence analysis was conducted to identify the conceptual structure and intellectual clusters within the ‘Dark Side of AI’ and ‘Consumer Creepiness’ literature. By utilizing the text-mining capabilities of VOSviewer (van Eck & Waltman, 2010), the titles and abstracts of the 56 sampled articles were analyzed. This process resulted in a network of 18 core thematic terms, interconnected through 55 links with a total link strength (TLS) of 79. As illustrated in Figure 3, the literature is organized into three distinct color-coded clusters that represent the psychological, decision-making, and strategic dimensions of the field.

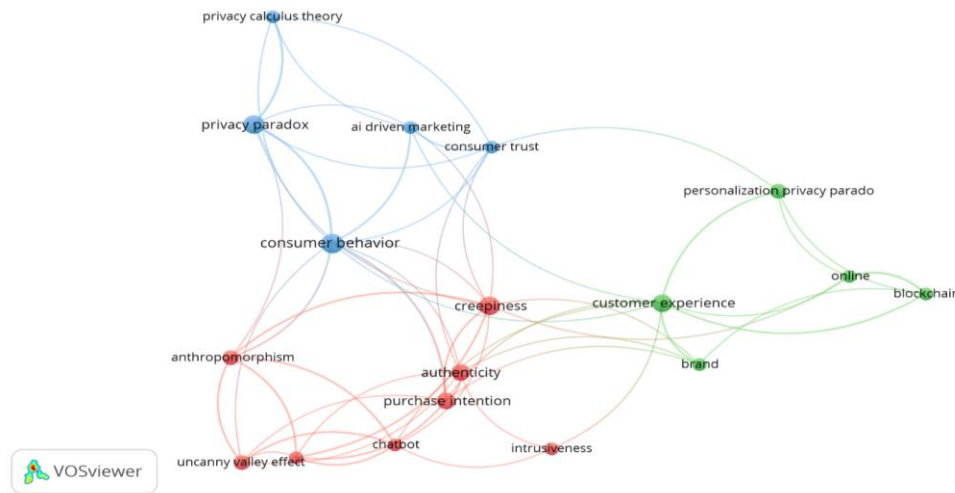


Figure 3. Co-occurrence network visualization of thematic terms in the “dark side of AI” and “consumer creepiness” literature

Visualization generated via VOSviewer using binary counting of terms in titles and abstracts ($n = 56$) (van Eck & Waltman, 2010). Node size indicates term frequency; link thickness represents co-occurrence strength. Cluster 1 (Red) identifies emotional and psychological drivers; Cluster 2 (Blue) represents the privacy-trust paradox; Cluster 3 (Green) highlights the strategic marketing context.

3.1.1 Cluster 1 (Red): “The uncanny valley & emotional drivers”

Includes: Creepiness, Intrusiveness, Anthropomorphism, Uncanny Valley Effect, Chatbot, Authenticity. This cluster represents the psychological core of the “Dark Side.” This cluster highlights how human-like AI (anthropomorphism) in chatbots and virtual influencers may trigger the Uncanny Valley Effect, leading to feelings of creepiness and reduced perceived authenticity. Recent evidence further suggests that uncanny perceptions may extend beyond immediate emotional discomfort to influence subsequent consumer decision-making and compensatory consumption behaviors, indicating that anthropomorphic AI can affect not only attitudes but also downstream purchasing responses (Hu, 2023).

The reviewed studies further indicate that anthropomorphic AI does not universally improve consumer evaluations. In hospitality settings, emphasizing robot-themed appeals may instead trigger unfavorable consumer responses when the AI characteristics become overly salient, suggesting that the effectiveness of anthropomorphic AI depends on consumers’ psychological interpretations rather than technological capability alone (Baek et al., 2025). Recent evidence further strengthens this interpretation by demonstrating that consumers’ responses to anthropomorphic products can be objectively observed through physiological indicators such as electroencephalography (EEG), skin conductance, and eye-tracking. These findings suggest that anthropomorphic AI influences consumers’ cognitive attention and emotional arousal beyond self-reported perceptions, providing additional support for the psychological mechanisms underlying the Intrusiveness Paradox (Xu et al., 2025).

This interpretation is further reinforced by recent evidence demonstrating that robotic anthropomorphism simultaneously enhances consumers’ social presence, enjoyment, and engagement while also increasing perceptions of creepiness and psychological discomfort. These dual effects support the central premise that human-like AI can generate both beneficial and adverse consumer responses, depending on how anthropomorphic technologies are designed and perceived (Mordi et al., 2026). This theoretical perspective is further supported by evidence from virtual influencer research, which demonstrates that the perceived human likeness of AI-generated

influencers simultaneously increases perceived authenticity and attractiveness while also elevating perceptions of eeriness, thereby shaping consumers attitudes and purchase intentions (Jang et al., 2023).

Recent chatbot research further demonstrates that anthropomorphic design can generate both positive and negative emotional responses. Specifically, chatbots with a higher experiential mind can evoke greater eeriness and amazement, illustrating how human-like characteristics may simultaneously produce psychological discomfort and positive emotional engagement in consumers (Seo & Yoon, 2025).

Recent evidence further suggests that perceived authenticity mediates the effectiveness of different types of influencers across product categories, emphasizing authenticity as a critical determinant of consumer acceptance of AI-generated and virtual influencers (Abdelsattar et al., 2024).

3.1.2 Cluster 2 (Blue): “The privacy-trust paradox”

Includes: Privacy Paradox, Privacy Calculus Theory, Consumer Trust, Consumer Behavior, AI-Driven Marketing. This cluster focuses on the decision-making process. It shows that even in AI-driven marketing, consumers weigh the benefits of personalization against their privacy concerns, reflecting the Privacy Paradox—the discrepancy between expressed privacy concerns and actual disclosure behavior. Recent evidence demonstrates that AI-driven consumer insights continue to influence purchasing behavior despite persistent privacy concerns (Alsiehem, 2025), while the personalization–privacy paradox literature further shows that consumers simultaneously value personalized experiences and remain concerned about the collection and use of their personal data (Duralia et al., 2025).

This tension represents a central mechanism through which AI-enabled personalization generates both perceived value and perceived intrusion, thereby reinforcing the foundation of the Intrusiveness Paradox proposed in this study.

3.1.3. Cluster 3 (Green): “The strategic marketing context”

Includes: Customer Experience, Personalization-Privacy Paradox, Brand, Online, Blockchain. This cluster connects the “Dark Side” to the bottom line. It shows how “Personalization” is the bridge between a positive Customer Experience and the potential for a privacy-related “Dark Side” backlash.

3.2 Contributing Countries

In the Table 2, top contributing countries in the niche topic of AI creepiness are presented.

Table 2. Top contributing countries in “AI creepiness” research ($n = 56$)

Country	Documents	Citations	TLS
United States	12	504	11
China	8	129	2
United Kingdom	4	28	3
Germany	4	4	1
Australia	3	488	6
Malaysia	3	457	7
India	3	456	3

NOTE: TLS = Total Link Strength; AI = Artificial Intelligence.

The United States demonstrated the highest TLS = 11, indicating its central position in the international collaboration network in the ‘Dark Side of AI’ discourse.

3.2.1 The global leaders (quantity vs. impact)

The United States is the undisputed leader in this niche, producing the highest volume of research (12 documents) and commanding the most academic influence with 504 citations. However, a “hidden gem” in the gathered data is Australia and Malaysia. They have fewer papers (3 each), and their citation-to-document ratios are exceptionally high (488 and 457 citations, respectively). This indicates that these specific papers are likely the “foundational” or “seminal” work that everyone else in the field is citing.

3.2.2 The collaboration hubs

The TLS indicates how much a country collaborates internationally.

The United States (TLS = 11) and Malaysia (TLS = 7) are the most “connected” countries.

This suggests that research on the “Dark Side of AI” is currently driven by Western-Eastern partnerships (e.g., US researchers collaborating with Malaysian and Indian scholars).

3.2.3 Emerging research hubs

China has a high output (8 documents), its TLS (2) is relatively low compared to the United State. This suggests that Chinese research in this field tends to be more “localized” or self-contained, whereas Western and Southeast Asian researchers are more globally integrated.

3.2.4 Interpreting global research patterns

Beyond publication counts, the geographical distribution of AI creepiness research appears to reflect differences in digital market maturity, regulatory environments, and societal attitudes toward privacy. The United States maintains leadership partly because of its highly developed digital economy and the early adoption of AI-driven marketing technologies by global platforms such as e-commerce providers, search engines, and social media companies. These conditions provide both extensive research funding opportunities and abundant real-world contexts for investigating AI-enabled personalization.

Australia and Malaysia demonstrate unusually high citation impact despite relatively small publication volumes. Citation counts and TLS values reflect the Dimensions.ai analysis conducted in February 2026. One possible explanation is that researchers in these countries have produced several influential studies addressing emerging ethical and privacy issues at an early stage of AI adoption. Their strong international collaboration networks may also increase research visibility and citation performance.

China’s relatively high publication output but lower TLS for international collaboration may reflect the concentration of research within domestic academic networks, differences in publication language, and the country’s distinct regulatory framework governing AI and digital platforms. Similarly, regional differences in privacy regulation, including stronger data protection regimes in some jurisdictions, may shape both research priorities and consumer responses toward AI-enabled personalization.

These findings suggest that AI intrusiveness is not solely a technological phenomenon but is also influenced by institutional, cultural, and regulatory contexts. Future comparative studies should investigate how national AI governance frameworks and cultural differences in privacy expectations influence consumer perceptions of algorithmic intrusiveness.

3.3 Co-Authorship Network

To understand the global collaborative landscape of research on AI creepiness, a co-authorship analysis by country was conducted. This visualization identifies the primary geographic hubs and the strength of international research partnerships. Using the VOSviewer software (van Eck & Waltman, 2010), the 56 identified publications were mapped to reveal the bibliometric connections between nations contributing to this discourse (Figure 4).



Figure 4. Co-authorship network of countries investigating AI creepiness ($n = 56$)

The visualization, generated via VOSviewer, displays the international collaborative ties between eight primary countries. The size of the nodes corresponds to the number of publications, while the thickness of the links represents the TLS of co-authored works. Two distinct clusters are identified: Cluster 1 (Red) representing a Western-Pacific Hub led by the United States, and Cluster 2 (Green) representing a Euro-African-Asian collaborative pathway. The geographical analysis reveals a globally dispersed yet interconnected research network consisting of 8 major countries organized into two primary clusters.

3.3.1 Cluster 1 (Red-western-pacific hub)

The United States acts as the central global “hub” for this research. Showing the strongest collaborative links with Australia, India, Malaysia, and South Korea. This suggests a robust cross-continental exchange between North American and Asia-Pacific researchers regarding the psychological impacts of AI.

3.3.2 Cluster 2 (Green-Euro-African-Asian hub)

This cluster indicates a distinct collaborative pathway between China, the United Kingdom, and Egypt. This

suggests that ethical concerns and AI creepiness are being investigated through both Western and Emerging Markets.

3.4 Key Findings

3.4.1 The United States dominance

The United States has the largest node, indicating it is currently the most prolific producer of research in this niche.

3.4.2 Strategic collaboration

The thick lines (links) between the United States and South Korea, and China and the United Kingdom, demonstrate that researchers are working across borders to understand how different cultures perceive AI intrusiveness.

3.4.3 Research gaps

Europe (except the United Kingdom) and South America are largely absent from this specific map of 56 papers.

4. Discussion

4.1 A Theory, Context, Characteristics, Methodology Synthesis of the Dark Side of AI in Consumer Behavior

Table 3. TCCM framework-dark side of AI in consumer behavior research ($n = 56$)

Theme	Theory	Context	Characteristics (Key Constructs)	Methodology	Representative Focus
AI personalization effectiveness vs intrusiveness	SOR; personalization theory	Online retail; AI-driven marketing	Perceived relevance; intrusiveness; trust; irritation	Quantitative (survey, SEM)	Personalization effectiveness vs intrusiveness
Virtual influencers & uncanny valley	Uncanny Valley Theory; social presence theory	Social media marketing	Human-likeness; eeriness; credibility; brand attitude	Experiments ; surveys	Consumer reactions to virtual influencers
AI ethics & moral evaluation	Ethical decision-making theory; moral psychology	AI-enabled digital marketing	Ethical concern; fairness; accountability	Conceptual; quantitative	Ethical judgment of AI marketing
Privacy paradox & data disclosure	Privacy calculus theory	Digital platforms; personalization systems	Privacy concern; perceived benefits; disclosure intention	Quantitative (SEM)	Data disclosure behavior in AI systems
Creepiness & psychological reactance	Psychological reactance theory	AI advertising; smart services	Creepiness; loss of control; avoidance	Experiments	Dark-side effects of AI targeting
Trust formation in AI	Trust transfer theory; TAM	Chatbots; recommender systems	Trust; perceived usefulness; perceived risk	Survey; SEM	Trust building in AI interactions
Chatbots & anthropomorphism	Human-computer interaction theory	Customer service automation	Anthropomorphism; satisfaction; continuance intention	Experiments ; surveys	Chatbot design and engagement
Algorithm aversion & control	Algorithm aversion theory; institutional trust	Automated decision systems	Control perception; resistance; acceptance	Mixed methods	Resistance to algorithmic decisions
Technostress & consumer well-being	Stressor-strain-outcome model	Digital advertising ecosystems	Fatigue; stress; ad avoidance	Quantitative	Well-being impacts of AI marketing
AI transparency & explainability	XAI framework	Algorithmic marketing systems	Transparency; explainability; trust	Conceptual; experiments	Role of AI transparency in trust

Note: TCCM = Theory, Context, Characteristics, Methodology; AI = Artificial Intelligence; SEM = Structural Equation Modeling; SOR = Stimulus-Organism-Response; TAM = Technology Acceptance Model; XAI = Explainable Artificial Intelligence.

The bibliometric analysis shows a concentrated yet globally diverse research landscape. The United States leads in research output with 12 studies, while Australia and Malaysia, despite producing fewer papers, demonstrate higher citation rates and have a major influence on AI creepiness research. Both countries also show strong

collaborative networks, as evidenced by high TLS scores that reflect active cross-country research partnerships. This review analyzed 56 peer-reviewed articles utilizing the TCCM framework, identifying key themes and research gaps. The field primarily relies on narrow range of theories, with most studies addressing limited contexts and employing similar methodologies (Table 3).

4.1.1 Theoretical concentration

Most studies rely on a small set of behavior theories. These include the Stimulus–Organism–Response (SOR) model, privacy calculus, and psychological reactance theory. They explain immediate and short-term reactions, such as discomfort and loss of trust. They also explain the avoidance of AI systems. However, they say little about power gaps or system control. They do not address long-term consumer freedom. Critical views, such as surveillance capitalism, are rare. Socio-technical system theory is also limited. The field needs broader and deeper use of theory.

4.1.2. Contextual dominance

Most studies focused on the context of digital marketing, social media, and online retail. Recent studies have also extended this research to AI-powered real-time recommendation systems in livestream commerce and tourism marketing, demonstrating that AI-driven recommendations influence customer engagement and purchase-related decisions while simultaneously raising questions about trust and personalization (Harris & Geer-Jeremiah, 2025). The reviewed studies further demonstrate that AI-enabled digital humans are increasingly used in livestream commerce, where consumers' repeat purchase intentions are influenced by perceptions of vividness, telepresence, credibility, and trust. These findings suggest that successful AI-enabled interactions depend not only on personalization capabilities but also on consumers' confidence in the authenticity and credibility of AI-mediated experiences (Wang et al., 2025). Many also examine social media settings, while recent studies have expanded this context to AI-enabled shopping technologies such as smart voice assistants, where consumers' adoption decisions are simultaneously influenced by perceived convenience, trust, and privacy concerns (Gelibolu & Mouloudj, 2025).

The reviewed literature also indicates that AI-driven retargeting has become an increasingly important application of personalized advertising, where consumer data are continuously analyzed to deliver highly tailored promotional messages. While such practices improve advertising relevance, they may simultaneously reinforce perceptions of persistent monitoring and psychological intrusion, further illustrating the personalization–intrusiveness tension identified in this review (Niekrasova & Kirnosova, 2025).

Recent evidence further demonstrates that AI-driven retargeting is becoming increasingly individualized through algorithmic name-personalized advertising, enabling organizations to tailor promotional messages at the individual consumer level. Although these approaches improve personalization and marketing effectiveness, they also intensify perceptions of continuous monitoring and personal surveillance, reinforcing the central Intrusiveness Paradox identified in this review (Vaidhyanathan, 2025).

The reviewed studies further indicate that AI increasingly supports consumer decision-making throughout the customer journey by combining personalized recommendations with user-generated content. While these capabilities can improve purchase decisions and customer experiences, they also increase consumers' awareness of continuous personalization and data-driven influence, reinforcing the broader personalization–intrusiveness tension identified in this review (Perst, 2026).

Recent evidence from social commerce further indicates that perceived personalization enhances customer experience and purchase intention; however, these positive effects may diminish when consumers experience similarity-induced confusion. This finding suggests that increasingly intensive personalization does not necessarily generate proportionally greater consumer value, reinforcing the review's conclusion that AI-enabled personalization produces both beneficial and adverse psychological outcomes depending on consumers' perceptions (Van My & Phong, 2025).

The reviewed studies demonstrate that AI applications have expanded across the advertising industry, supporting audience targeting, personalized advertising, engagement optimization, predictive analytics, and campaign performance evaluation. These developments illustrate how AI increasingly shapes multiple stages of marketing communication while simultaneously expanding opportunities for extensive consumer data collection and personalized interactions (Ryzhko et al., 2024).

The reviewed literature also highlights social media marketing as a major application of AI, where chatbots, personalized content, interactive communication, and AI-enabled community building enhance customer experiences and engagement. However, these benefits continue to coexist with consumer concerns regarding privacy, trust, and perceived intrusiveness, reinforcing the central personalization–intrusiveness tension synthesized in this review (Arpitha et al., 2025). Similarly, the reviewed studies indicate that AI is transforming social media marketing by enabling more personalized customer engagement, content optimization, and marketing effectiveness. At the same time, these technological advances intensify concerns regarding privacy, transparency, algorithmic bias, and ethical AI use, further illustrating the coexistence of AI-enabled benefits and consumer risks

across digital marketing environments (Sharma & Sharma, 2024).

Although digital marketing remains the dominant research context, the reviewed literature also demonstrates growing scholarly interest in AI-related ethical challenges beyond conventional marketing environments. For example, studies conducted in healthcare-related digital ecosystems identify similar concerns regarding privacy, transparency, trust, and responsible AI adoption, suggesting that the adverse psychological and ethical consequences of AI-enabled interactions extend across multiple application domains (Bhatt & Jhavar, 2025).

The research mainly comes from Western countries. Other regions receive far less attention. Emerging markets are rarely studied. Cross-cultural differences are often ignored. Non-business areas, such as health and education, are limited. Nevertheless, recent evidence suggests that similar concerns regarding trust, privacy, and perceived creepiness also influence consumer adoption of AI in healthcare, indicating that the mechanisms underlying the Intrusiveness Paradox may extend beyond commercial marketing settings into other AI-enabled service domains (Kumar et al., 2025). Public service uses of AI are also underexplored. This narrow viewpoint limits broader insight. Cultural views on AI ethics and intrusiveness need more study.

4.1.3. Characteristics emphasis

Most studies looked at users' trust, feelings of creepiness, privacy concerns, perceived control, and ethical judgment. These variables primarily capture short-term cognitive and emotional effects, explaining immediate reactions to AI systems. Recent evidence conceptualizes creepiness as a distinct emotional response emerging when AI-enabled personalized marketing is simultaneously perceived as ambiguous and intrusively surveilling. Such emotional responses subsequently trigger consumer reactance and reduce purchase intentions, highlighting creepiness as a central mechanism through which AI-enabled personalization may produce adverse consumer outcomes (Petrova et al., 2025). Recent research further suggests that technologically advanced data collection practices may be perceived as psychologically intrusive when consumers believe organizations observe, infer, or anticipate personal behaviors beyond acceptable social boundaries, thereby intensifying concerns regarding privacy, trust, and ethical acceptability (Krause & Groeppel-Klein, 2026).

Privacy expectations may also vary according to the communication modality through which consumers interact with AI. Recent evidence suggests that conversational interactions, particularly voice-based communication, often elicit stronger expectations of privacy than text-based interactions, indicating that consumer responses to AI depend not only on data collection practices but also on the mode of human–AI communication (Melzner et al., 2026). However, few studies have tracked longitudinal outcomes. Habit formation and consumer resilience over time are rarely examined, and digital well-being receives limited attention. The field has largely focused on micro-level psychological biases, while broader social and market effects call for further investigation.

4.1.4. Methodological conservatism

Most studies use cross-sectional surveys and lab experiments. Many rely on structural equation modeling (SEM). These methods are strong but limited. They do not show long-term cause and effect.

They also lack real-life settings. Longitudinal studies are scarce. Field experiments are limited.

Qualitative research is uncommon. Mixed-method designs are also scarce. This limits deeper insight into developing consumer–AI relationships.

4.1.5 Integrating Theory, Context, Characteristics and Methods

The TCCM analysis reveals that the four dimensions are closely interconnected rather than independent components of literature. Different theoretical perspectives tend to dominate specific research contexts and explain different aspects of consumer responses toward AI.

Privacy Calculus Theory is primarily employed in studies examining disclosure decisions, perceived benefits, and trust formation within recommendation systems, personalized advertising, and digital platforms. These studies predominantly utilize survey-based quantitative methods and SEM, emphasizing rational evaluations of benefits and privacy risks. At the same time, the reviewed literature highlights that consumer responses are also shaped by broader ethical concerns, including privacy protection, algorithmic bias, transparency, and responsible AI governance, which increasingly influence trust in AI-enabled marketing systems (Clement et al., 2025). Recent systematic review evidence further indicates that AI-powered recommender systems have evolved toward more adaptive, privacy-conscious, and trust-oriented designs, highlighting that effective personalization increasingly depends on balancing accuracy with consumer expectations regarding transparency, privacy protection, and user engagement (Kumari & Laheri, 2025).

By contrast, Uncanny Valley Theory is most frequently applied in studies investigating anthropomorphic chatbots, virtual influencers, and AI-generated avatars. Because these studies focus on emotional responses such as discomfort, eeriness, and perceived authenticity, experimental research designs are considerably more common than surveys.

Psychological Reactance Theory occupies an intermediate position by explaining behavioral resistance following perceived autonomy loss. Research grounded in this perspective often combines controlled experiments

with behavioral intention measures, highlighting the transition from cognitive perceptions of control toward actual resistance behaviors.

These patterns demonstrate that methodological choices are closely aligned with theoretical assumptions. Studies emphasizing cognitive decision-making generally rely on cross-sectional surveys, whereas investigations of emotional reactions increasingly adopt experimental approaches. Nevertheless, longitudinal and mixed-method studies remain scarce across all theoretical traditions, limiting current understanding of how AI-related trust, discomfort, and resistance evolve over time.

Collectively, these findings demonstrate that the Intrusiveness Paradox emerges from the interaction of emotional, cognitive, and behavioral mechanisms rather than any single theoretical perspective. The TCCM synthesis therefore supports the need for an integrative framework capable of connecting these previously fragmented research streams.

4.2 Research Gaps Identified Through the Theory, Context, Characteristics, Methodology Framework

4.2.1 Theory gaps

- Over-reliance on SOR and privacy calculus models
- Lack of critical, institutional, and power-based theories
- Minimal theorization of long-term consumer autonomy

4.2.2. Context gaps

- Limited studies in emerging economies and non-Western cultures
- Underexplored non-commercial AI applications
- Absence of cross-cultural comparative designs

4.2.3. Characteristic gaps

- Focus on short-term reactions over long-term outcomes
- Limited integration of consumer well-being and resilience
- Neglect of firm-level and societal consequences

4.2.4. Methodological gaps

- Scarcity of longitudinal and field-based research
- Minimal qualitative and interpretive approaches
- Limited use of real AI systems instead of hypothetical scenarios

4.3 Suggested Future Research Directions

- Develop multi-level frameworks that link consumer psychology to algorithmic governance
- Additional research could expand to European, West Asian, and Latin American countries to see if ‘creepiness’ is viewed differently in those countries
- Conduct cross-cultural longitudinal studies on AI trust erosion and recovery
- Apply mixed-method and field experiments using real AI interfaces
- Integrate consumer well-being and digital ethics as core outcome variables

5. Conclusions

This study develops and theoretically positions the Intrusiveness Paradox as an integrative framework that explains the simultaneous coexistence of personalization benefits and perceived psychological intrusion arising from AI-enabled marketing. Drawing on evidence synthesized from 56 peer-reviewed studies, the review demonstrates that AI-driven personalization generates not only functional value but also privacy concerns, perceived surveillance, emotional discomfort, diminished trust, and consumer resistance. The reviewed evidence also confirms that organizations continue to invest in AI-enabled personalization because it enhances marketing performance, customer targeting, and customer engagement across digital commerce environments. These organizational benefits explain why AI-driven personalization continues to expand despite the psychological and privacy-related concerns identified throughout the reviewed literature, reinforcing the dual nature of the Intrusiveness Paradox. By integrating these previously fragmented research streams, the proposed framework extends existing theories—including Privacy Calculus Theory, the Privacy Paradox, Uncanny Valley Theory, and Psychological Reactance Theory—and provides a more comprehensive explanation of consumers’ responses to AI-enabled personalization. This theoretical advancement also extends the personalization–privacy paradox by demonstrating that consumers’ responses to AI-enabled personalization are influenced not only by privacy-related cost–benefit evaluations but also by perceptions of psychological intrusion, emotional discomfort, and consumer

reactance. In this way, the Intrusiveness Paradox broadens existing explanations of AI-enabled personalization by incorporating psychological mechanisms that extend beyond data disclosure decisions.

5.1 Theoretical Contributions

This review integrates findings from 56 peer-reviewed studies on the dark side of AI, with particular emphasis on AI and consumer decision processes. Although creepiness, intrusiveness, and privacy issues are often examined as separate constructs, the evidence indicates that these occurrences are closely interconnected. Together, they reflect a shift in the conceptualization of human–AI technology interaction. Creepiness is not simply an extension of privacy problems; instead, it represents basic structural tensions identified in Uncanny Valley Theory and Social Presence Theory. As AI systems become more human-like, initial user involvement generally increases. After a certain point, people start to feel uncomfortable. This review uses the TCCM framework to explain the “Intrusiveness Paradox.” While AI makes personalization and customer experience better, it also creates worries about losing control and being watched. The framework extends existing perspectives such as the Privacy Paradox by integrating emotional, cognitive, and behavioral mechanisms that jointly explain consumer responses to AI-enabled personalization. This interpretation is consistent with recent evidence demonstrating that generative AI simultaneously creates organizational and consumer value while introducing challenges related to trust, transparency, and ethical concerns, reinforcing the need to understand both the positive and negative consequences of AI-enabled interactions within an integrated conceptual framework. This interpretation is also consistent with recent research on phygital customer experiences, which argues that generative AI simultaneously enhances customer value through personalization and seamless interactions while introducing new challenges related to privacy, transparency, trust, and ethical governance, reinforcing the need to balance AI-enabled benefits with potential psychological and ethical risks.

The proposed framework therefore provides a broader explanation of how consumers simultaneously appreciate AI-enabled personalization while experiencing psychological discomfort and perceived intrusion. This distinction is further reinforced by recent systematic review evidence on the AI-driven personalization–privacy paradox, which primarily conceptualizes the tension as a trade-off between personalization benefits and privacy concerns. Building upon this perspective, the Intrusiveness Paradox proposed in this study extends beyond privacy-related decision-making by incorporating psychological intrusion, perceived creepiness, emotional discomfort, and consumer reactance into a broader integrative framework for explaining consumer responses to AI-enabled marketing. The Stimulus–Organism–Response model and Privacy Calculus Theory help explain how people react in the short term, but they do not fully cover the long-term effects of close contact with AI. This study brings together emotion, thinking, and trust in one model. It shows that AI risk is a key issue in marketing theory. The negative side of AI is not a small problem for consumers and marketers. It is a major challenge for marketing research today.

5.2 Managerial Implications

5.2.1 Recommendation systems

Organizations using AI recommendation systems should balance personalization with perceived autonomy. Allowing users to adjust recommendation settings, review personalization preferences, or temporarily disable algorithmic profiling may reduce perceptions of surveillance while maintaining personalization effectiveness. As generative AI increasingly provides personalized recommendations, organizations should also ensure that AI-generated advice is accurate, explainable, and presented as decision support rather than a substitute for consumer judgment. Maintaining transparency and allowing consumers to critically evaluate AI recommendations can strengthen trust and reduce resistance toward AI-assisted decision-making.

5.2.2 Chatbots

Chatbots should be designed to achieve an optimal level of anthropomorphism. Human-like conversational interfaces can improve engagement, but excessive realism may trigger the Uncanny Valley Effect and increase consumer discomfort. Designers should therefore prioritize transparency by clearly identifying AI agents and avoiding deceptive human imitation. Similarly, organizations deploying AI-enabled voice agents should recognize that successful service interactions depend not only on functional performance but also on consumers’ perceptions of trust, conversational naturalness, and human-like interaction, all of which influence acceptance and continued usage.

5.2.3 Personalized advertising

Personalized advertising should emphasize transparency regarding data collection and personalization logic. XAI features, privacy dashboards, and granular consent mechanisms may strengthen consumer trust while reducing feelings of algorithmic intrusion.

5.2.4 AI governance

Organizations should complement technical solutions with organizational governance mechanisms, including ethical AI audits, human oversight, algorithmic accountability, and regular privacy impact assessments. These practices help preserve long-term customer trust while reducing reputational risks associated with intrusive AI applications. The reviewed literature further suggests that organizations should complement these practices by embedding fairness, transparency, and accountability into AI systems throughout their design and deployment. Implementing fairness-oriented AI governance can reduce algorithmic bias, strengthen consumer trust, and support more responsible AI-enabled marketing practices, thereby mitigating perceptions of psychological intrusion.

5.2.5 Digital platforms

Managers operating e-commerce platforms and digital marketplaces should continuously monitor indicators such as customer trust, perceived creepiness, privacy concerns, and opt-out behavior as strategic performance metrics rather than treating personalization solely as a marketing optimization tool.

5.3 Limitations

The following are limitations of the study:

First, the search relied only on Dimensions.ai. Other databases, such as Web of Science and Scopus, were not included. Some relevant studies may therefore have been missed.

Second, the review included 56 peer-reviewed journal articles. Conference papers, industry white papers, and practitioner reports were excluded. This ensured rigor but narrowed the scope.

Third, most studies were conducted in Western and East Asian countries. Europe, West Asia, Africa, and Latin America were less represented. Cultural differences in AI privacy may not be fully captured.

Finally, many studies used quantitative survey methods. Longitudinal consumer-AI relationship studies were limited. This limits insight into the development of consumer-AI relationships over time.

5.4 Future Research Agenda

Based on the identified TCCM gaps, future studies should use more diverse methods. Longitudinal design studies can track AI aversion over time. Does AI discomfort grow or fade over time? Neuromarketing approaches and behavioral experiments can uncover subconscious and unconscious responses to anthropomorphic cues. Lab and field tests with real AI tools are needed. These designs show behavior, not just self-reports. Furthermore, enriching theoretical development requires mixed-methods research that combines qualitative interviews with big-data analytics, which can add depth to survey results.

Other sectors, such as healthcare, fintech, and public services, need attention and further studies. Risks are higher in these sectors. Creepiness and privacy failures/breaches can cause major ethical and social consequences. Cross-cultural studies are essential. Views on intrusiveness may vary across regulatory, social, and cultural systems, and researchers must take this into account, as rules and norms shape consumer sentiment. AI is becoming part of daily commerce and marketing research. Research must move beyond better targeting. It must also protect consumer autonomy. The most important concern for the next decade is not how AI will develop, but rather how it should be built to preserve trust in an era of algorithmic intimacy.

Author Contributions

Conceptualization, S.A.F. and J.C.P.; methodology, S.A.F.; software, S.A.F.; validation, J.C.P.; formal analysis, S.A.F.; investigation, J.C.P.; resources, J.C.P.; data curation, J.C.P.; writing—original draft preparation, S.A.F.; writing—review and editing, J.C.P.; visualization, S.A.F.; supervision, J.C.P.; project administration, S.A.F.; funding acquisition, J.C.P. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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